day_8_genome_assembly_Debarshi

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```
library(Rsubread)
setwd("/home/ibab/bio-env")
gtf_file <- "GCF_000146045.2_R64_genomic.gtf"

out_FeatCount_conA_rep1 <- featureCounts(
  files = "conA_rep1.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)
```

```
##
##          ==========     _____ _     _ ____  _____  _______          _____
##          =====         / ____| |  | |  _ \|  __ \|  ____|   /\    |  __ \
##            =====      | (___ | |  | | |_) | |__) | |__     /  \   | |  | |
##              ====      \___ \| |  | |  _ <|  _  /|  __|   / /\ \  | |  | |
##                 ====   ____) | |__| | |_) | | \ \| |____ / ____ \| |__| |
##          ==========   |_____/ \____/|____/|_|  \_\______/_/    \_\_____/
##         Rsubread 2.24.0
##
## //========================== featureCounts setting ===========================\\
## ||                                                                            ||
## ||             Input files : 1 SAM file                                       ||
## ||                                                                            ||
## ||                           conA_rep1.sam                                    ||
## ||                                                                            ||
## ||              Paired-end : no                                               ||
## ||        Count read pairs : no                                               ||
## ||              Annotation : GCF_000146045.2_R64_genomic.gtf (GTF)            ||
## ||      Dir for temp files : .                                                ||
## ||                 Threads : 1                                                ||
## ||                   Level : meta-feature level                               ||
## ||      Multimapping reads : counted                                          ||
## || Multi-overlapping reads : not counted                                      ||
## ||   Min overlapping bases : 1                                                ||
## ||                                                                            ||
## \\============================================================================//
##
## //================================ Running ===================================\\
## ||                                                                            ||
## || Load annotation file GCF_000146045.2_R64_genomic.gtf ...                   ||
## ||    Features : 6831                                                         ||
## ||    Meta-features : 6459                                                    ||
## ||    Chromosomes/contigs : 17                                                ||
```

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```
## ||                                                                            ||
## || Process SAM file conA_rep1.sam...                                          ||
## ||    Single-end reads are included.                                          ||
## ||    Total alignments : 800000                                               ||
## ||    Successfully assigned alignments : 721531 (90.2%)                       ||
## ||    Running time : 0.01 minutes                                             ||
## ||                                                                            ||
## || Write the final count table.                                               ||
## || Write the read assignment summary.                                         ||
## ||                                                                            ||
## \\============================================================================//
```

```
out_FeatCount_conA_rep2 <- featureCounts(
  files = "conA_rep2.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)
```

```
##
##          ==========     _____ _     _ ____  _____  _______          _____
##          =====         / ____| |  | |  _ \|  __ \|  ____|   /\    |  __ \
##            =====      | (___ | |  | | |_) | |__) | |__     /  \   | |  | |
##              ====      \___ \| |  | |  _ <|  _  /|  __|   / /\ \  | |  | |
##                ====    ____) | |__| | |_) | | \ \| |____ / ____ \| |__| |
##          ==========   |_____/ \____/|____/|_|  \_\______/_/    \_\_____/
##         Rsubread 2.24.0
##
## //========================== featureCounts setting ===========================\\
## ||                                                                            ||
## ||             Input files : 1 SAM file                                       ||
## ||                                                                            ||
## ||                           conA_rep2.sam                                    ||
## ||                                                                            ||
## ||              Paired-end : no                                               ||
## ||        Count read pairs : no                                               ||
## ||              Annotation : GCF_000146045.2_R64_genomic.gtf (GTF)            ||
## ||      Dir for temp files : .                                                ||
## ||                 Threads : 1                                                ||
## ||                   Level : meta-feature level                               ||
## ||      Multimapping reads : counted                                          ||
## || Multi-overlapping reads : not counted                                      ||
## ||   Min overlapping bases : 1                                               ||
## ||                                                                            ||
## \\============================================================================//
##
## //================================= Running ==================================\\
## ||                                                                            ||
## || Load annotation file GCF_000146045.2_R64_genomic.gtf ...                   ||
## ||    Features : 6831                                                         ||
## ||    Meta-features : 6459                                                    ||
## ||    Chromosomes/contigs : 17                                                ||
```

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```
## ||                                                                            ||
## || Process SAM file conA_rep2.sam...                                          ||
## ||    Single-end reads are included.                                          ||
## ||    Total alignments : 900000                                               ||
## ||    Successfully assigned alignments : 816269 (90.7%)                       ||
## ||    Running time : 0.01 minutes                                             ||
## ||                                                                            ||
## || Write the final count table.                                               ||
## || Write the read assignment summary.                                         ||
## ||                                                                            ||
## \\============================================================================//
```

```
out_FeatCount_conB_rep1 <- featureCounts(
  files = "conB_rep1.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)
```

```
##
##          ==========     _____ _     _ ____  _____  _______          _____
##          =====         / ____| |  | |  _ \|  __ \|  ____|   /\    |  __ \
##            =====      | (___ | |  | | |_) | |__) | |__     /  \   | |  | |
##              ====      \___ \| |  | |  _ <|  _  /|  __|   / /\ \  | |  | |
##                ====    ____) | |__| | |_) | | \ \| |____ / ____ \| |__| |
##          ==========   |_____/ \____/|____/|_|  \_\______/_/    \_\_____/
##         Rsubread 2.24.0
##
## //========================== featureCounts setting ===========================\\
## ||                                                                            ||
## ||             Input files : 1 SAM file                                       ||
## ||                                                                            ||
## ||                           conB_rep1.sam                                    ||
## ||                                                                            ||
## ||              Paired-end : no                                               ||
## ||        Count read pairs : no                                               ||
## ||              Annotation : GCF_000146045.2_R64_genomic.gtf (GTF)            ||
## ||      Dir for temp files : .                                                ||
## ||                 Threads : 1                                                ||
## ||                   Level : meta-feature level                               ||
## ||      Multimapping reads : counted                                          ||
## || Multi-overlapping reads : not counted                                      ||
## ||   Min overlapping bases : 1                                               ||
## ||                                                                            ||
## \\============================================================================//
##
## //================================ Running ===================================\\
## ||                                                                            ||
## || Load annotation file GCF_000146045.2_R64_genomic.gtf ...                   ||
## ||    Features : 6831                                                         ||
## ||    Meta-features : 6459                                                    ||
## ||    Chromosomes/contigs : 17                                                ||
```

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```
## ||                                                                            ||
## || Process SAM file conB_rep1.sam...                                          ||
## ||    Single-end reads are included.                                          ||
## ||    Total alignments : 700000                                               ||
## ||    Successfully assigned alignments : 633555 (90.5%)                       ||
## ||    Running time : 0.01 minutes                                             ||
## ||                                                                            ||
## || Write the final count table.                                               ||
## || Write the read assignment summary.                                         ||
## ||                                                                            ||
## \\============================================================================//
```

```
out_FeatCount_conB_rep2 <- featureCounts(
  files = "conB_rep2.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)
```

```
##
##          ==========     _____ _     _ ____  _____  _______          _____
##          =====         / ____| |  | |  _ \|  __ \|  ____|   /\    |  __ \
##            =====      | (___ | |  | | |_) | |__) | |__     /  \   | |  | |
##              ====      \___ \| |  | |  _ <|  _  /|  __|   / /\ \  | |  | |
##                ====    ____) | |__| | |_) | | \ \| |____ / ____ \| |__| |
##          ==========   |_____/ \____/|____/|_|  \_\______/_/    \_\_____/
##         Rsubread 2.24.0
##
## //========================== featureCounts setting ===========================\\
## ||                                                                            ||
## ||             Input files : 1 SAM file                                       ||
## ||                                                                            ||
## ||                           conB_rep2.sam                                    ||
## ||                                                                            ||
## ||              Paired-end : no                                               ||
## ||        Count read pairs : no                                               ||
## ||              Annotation : GCF_000146045.2_R64_genomic.gtf (GTF)            ||
## ||      Dir for temp files : .                                                ||
## ||                 Threads : 1                                                ||
## ||                   Level : meta-feature level                               ||
## ||      Multimapping reads : counted                                          ||
## || Multi-overlapping reads : not counted                                      ||
## ||   Min overlapping bases : 1                                               ||
## ||                                                                            ||
## \\============================================================================//
##
## //================================ Running ===================================\\
## ||                                                                            ||
## || Load annotation file GCF_000146045.2_R64_genomic.gtf ...                   ||
## ||    Features : 6831                                                         ||
## ||    Meta-features : 6459                                                    ||
## ||    Chromosomes/contigs : 17                                                ||
```

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```
## ||                                                                      ||
## || Process SAM file conB_rep2.sam...                                    ||
## ||    Single-end reads are included.                                    ||
## ||    Total alignments : 1000000                                        ||
## ||    Successfully assigned alignments : 903130 (90.3%)                 ||
## ||    Running time : 0.01 minutes                                       ||
## ||                                                                      ||
## || Write the final count table.                                         ||
## || Write the read assignment summary.                                   ||
## ||                                                                      ||
## \\======================================================================//
```

```
out_FeatCount_conA_rep1_count = data.frame(out_FeatCount_conA_rep1$counts)
out_FeatCount_conA_rep2_count = data.frame(out_FeatCount_conA_rep2$counts)
out_FeatCount_conB_rep1_count = data.frame(out_FeatCount_conB_rep1$counts)
out_FeatCount_conB_rep2_count = data.frame(out_FeatCount_conB_rep2$counts)
```

```
count_matrix <- cbind(
out_FeatCount_conA_rep1_count,
out_FeatCount_conA_rep2_count,
out_FeatCount_conB_rep1_count,
out_FeatCount_conB_rep2_count
)
write.table(count_matrix, file = "count_matrix.txt",
sep="\t")
```

```
library(edgeR)
```

```
## Warning: package 'edgeR' was built under R version 4.5.3
```

```
## Loading required package: limma
```

```
## Loading required package: limma
group=c(rep("conA",2),rep("conB",2))
```

```
y=DGEList(counts=count_matrix,group=group)
y
```

```
## An object of class "DGEList"
## $counts
##           conA_rep1.sam conA_rep2.sam conB_rep1.sam conB_rep2.sam
## YAL068C               0             0             0             0
## YAL067W-A             0             0             0             0
## YAL067C               0             0             0             0
## YAL065C               0             0             0             0
## YAL064W-B             1             0             1             0
## 6454 more rows ...
##
```

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```
## $samples
##                group lib.size norm.factors
## conA_rep1.sam  conA   721531            1
## conA_rep2.sam  conA   816269            1
## conB_rep1.sam  conB   633555            1
## conB_rep2.sam  conB   903130            1
```

```
# Filter lowly expressed genes #filterByExpr() identifies genes with sufficient express
ion to be reliably tested. Applies the cpm and gives cutoff
keep=filterByExpr(y)
head(keep)
```

```
##   YAL068C YAL067W-A   YAL067C   YAL065C YAL064W-B YAL064C-A
##     FALSE     FALSE     FALSE     FALSE     FALSE     FALSE
```

```
# how many genes survived
sum(keep)
```

```
## [1] 5224
```

```
# to see the total number of genes before filtering.
length(keep)
```

```
## [1] 6459
```

```
library(edgeR)

# 1. Ensure design matrix is defined first
design1 <- model.matrix(~0 + group)

# 2. Recalculate 'keep' using filterByExpr on object 'y'
keep <- filterByExpr(y, design = design1)

# 3. Filter the DGEList (lengths will now match)
y <- y[keep, , keep.lib.sizes = FALSE]
dim(y)
```

```
## [1] 5224    4
```

```
# 4. Calculate normalization factors (TMM)
y <- calcNormFactors(y)

# 5. Estimate dispersion
y <- estimateDisp(y, design1)

# 6. Test for differential expression
# Note: exactTest requires treatment group factors; ensure group is set on y$samples$gr
oup
```

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```
DEet <- exactTest(y, pair = c("conA", "conB"))

# View top results
head(DEet$table)
```

```
##              logFC   logCPM     PValue
## YAL063C -0.7996891 3.871542 0.17528734
## YAL062W -0.7794297 4.791989 0.10740418
## YAL061W -0.8961737 8.968509 0.04103521
## YAL060W -0.7463570 9.689610 0.04324005
## YAL059W  1.0540346 4.477723 0.05977212
## YAL058W  0.4713354 5.690378 0.13805991
```

```
#Identifying the up and down regulated gene at least 2-fold higher expression and stron
g statistical significance
et_table = DEet$table
up_et = subset(et_table,et_table$logFC >= 1 & et_table$PValue <= 0.001)
down_et = subset(et_table,et_table$logFC <= -1 & et_table$PValue <= 0.001)
```

```
# up regulated gene
head(up_et)
```

```
##            logFC    logCPM       PValue
## YAL025C 2.725719  5.599999 3.113077e-13
## YBL042C 1.930598  6.457424 2.155048e-11
## YBL039C 1.959675  6.764031 8.795183e-08
## YBL024W 1.488132  6.681218 1.595984e-04
## YBL004W 1.130756  6.618267 4.097396e-04
## YBR031W 1.453359 11.303596 9.933219e-04
```

```
# Downregulated genes
head(down_et)
```

```
##               logFC   logCPM       PValue
## YAL008W   -1.311142 6.210347 4.367562e-06
## YBL064C   -1.235612 8.116882 2.765428e-04
## YBL015W   -1.400856 8.964937 1.640541e-04
## YBR053C   -1.087734 7.310590 9.051842e-04
## YBR072W   -3.510040 8.674449 2.186622e-12
## YBR089C-A -1.232368 5.891167 4.559913e-05
```

```
#No of gene upregulated and down regulated
nrow(up_et)
```

```
## [1] 235
```

```
nrow(down_et)
```

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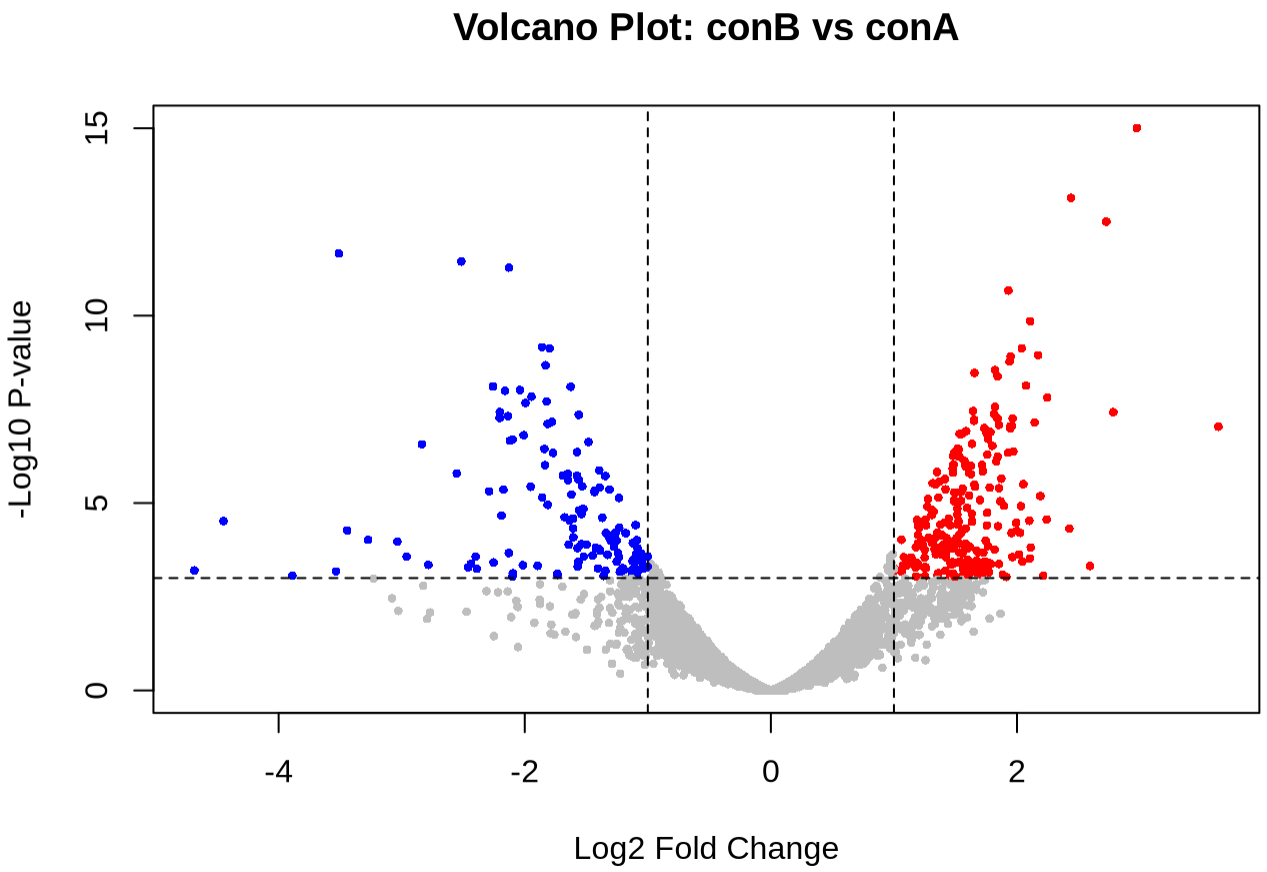
```
## [1] 135
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
# Create a color column
et_table$color <- "grey"
# Upregulated genes
et_table$color[et_table$logFC >= 1 & et_table$PValue <= 0.001] <- "red"
# Downregulated genes
et_table$color[et_table$logFC <= -1 & et_table$PValue <= 0.001] <- "blue"
# Calculate -log10(P-value)
et_table$negLogP <- -log10(et_table$PValue)
# Volcano plot
plot(
et_table$logFC,
et_table$negLogP,
pch = 16,
cex = 0.6,
col = et_table$color,
xlab = "Log2 Fold Change",
ylab = "-Log10 P-value",
main = "Volcano Plot: conB vs conA"
)
# Vertical cutoff lines
abline(
v = c(-1, 1),
lty = 2,
col = "black"
)
# Horizontal cutoff line
abline(
h = -log10(0.001),
lty = 2,
col = "black"
)
```

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```
# RPKM_Normalization for raw read counts
gene_length=out_FeatCount_conA_rep1$annotation$Length
head(gene_length)
```

```
## [1]  363  228 1782  387  381  381
```

```
gene_id <- out_FeatCount_conA_rep1$annotation$GeneID
head(gene_id)
```

```
## [1] "YAL068C"   "YAL067W-A" "YAL067C"   "YAL065C"   "YAL064W-B" "YAL064C-A"
```

```
gene_info <- data.frame(Geneid = gene_id,Length = gene_length)
head(gene_info)
```

```
##      Geneid Length
## 1   YAL068C    363
## 2 YAL067W-A    228
## 3   YAL067C   1782
## 4   YAL065C    387
## 5 YAL064W-B    381
## 6 YAL064C-A    381
```

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```
gene_info <- gene_info[match(rownames(count_matrix), gene_info$Geneid),]
count_data <- cbind(
Geneid = rownames(count_matrix),
Length = gene_info$Length,
count_matrix,
row.names = NULL
)
write.table(count_data, file = "count_data.txt",
sep="\t")
dim(count_data)
```

```
## [1] 6459    6
```

```
library_size <- colSums(count_data[, 3:6])
library_size
```

```
## conA_rep1.sam conA_rep2.sam conB_rep1.sam conB_rep2.sam
##        721531        816269        633555        903130
```

```
scaling_factor <- library_size / 1000000
scaling_factor
```

```
## conA_rep1.sam conA_rep2.sam conB_rep1.sam conB_rep2.sam
##      0.721531      0.816269      0.633555      0.903130
```

```
gene_length_kb <- count_data$Length / 1000
#Calculate RPKM
RPKM_conA_rep1<-count_data$conA_rep1.sam/(gene_length_kb*scaling_factor[1])
RPKM_conA_rep2<-count_data$conA_rep2.sam/(gene_length_kb*scaling_factor[2])
RPKM_conB_rep1<-count_data$conB_rep1.sam/(gene_length_kb*scaling_factor[3])
RPKM_conB_rep2<-count_data$conB_rep2.sam/(gene_length_kb*scaling_factor[4])
#Create the RPKM matrix
rpkm <- data.frame(
Geneid = count_data$Geneid,
RPKM_conA_rep1 = RPKM_conA_rep1,
RPKM_conA_rep2 = RPKM_conA_rep2,
RPKM_conB_rep1 = RPKM_conB_rep1,
RPKM_conB_rep2 = RPKM_conB_rep2
)
rownames(rpkm) <- rpkm$Geneid
rpkm$Geneid <- NULL
#Log transform for heatmap
log_rpkm <- log2(rpkm + 1)
```

```
#For DEG list perform RPKM Normalization
up_genes <- rownames(up_et)
down_genes <- rownames(down_et)
```

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```
deg_genes <- c(up_genes, down_genes)
length(deg_genes)
```

```
## [1] 370
```

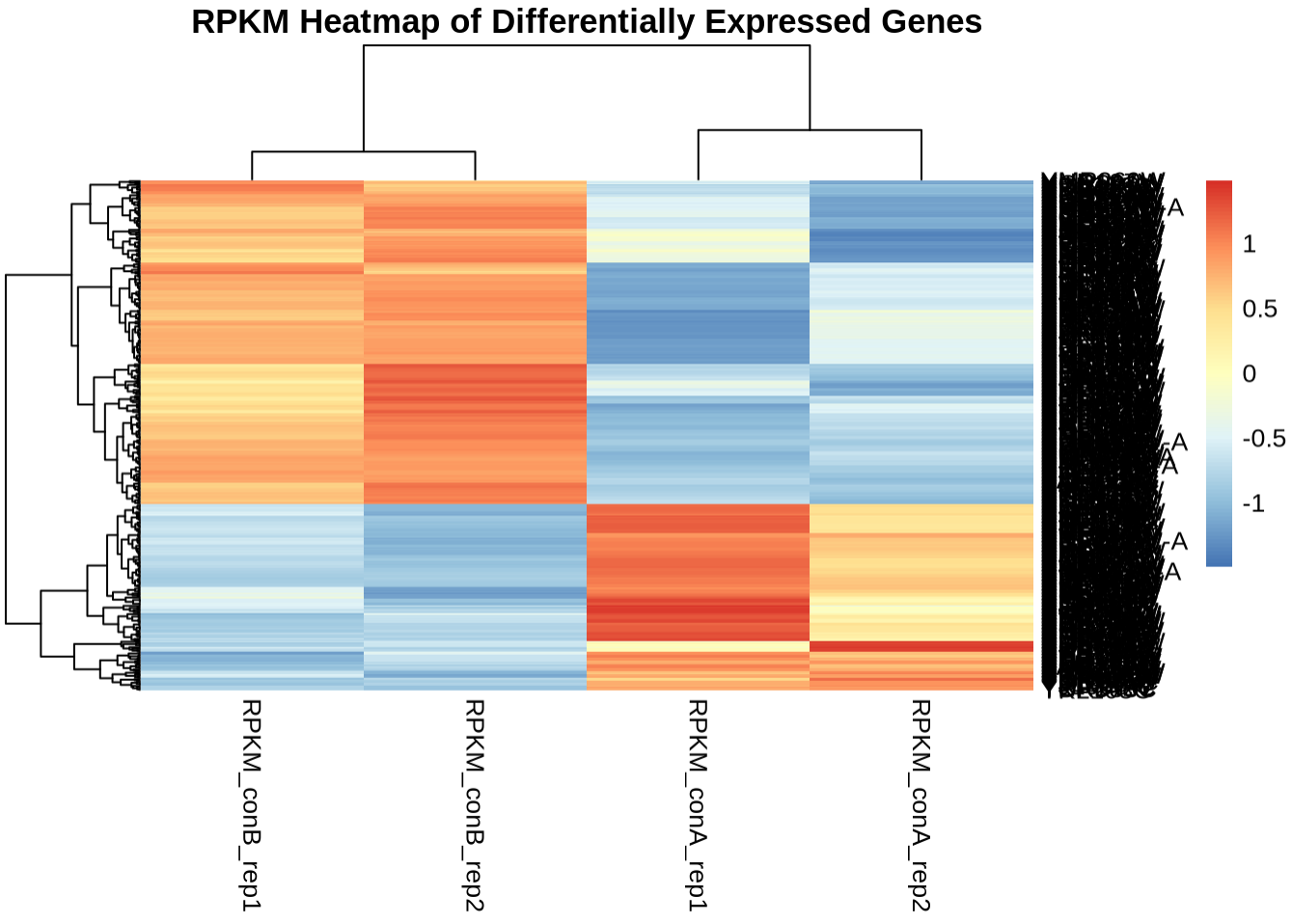
```
rpkm_deg <- rpkm[deg_genes, ]
dim(rpkm_deg)
```

```
## [1] 370   4
```

```
log_rpkm_deg <- log2(rpkm_deg + 1)
library(pheatmap)
```

```
## Warning: package 'pheatmap' was built under R version 4.5.3
```

```
pheatmap(
log_rpkm_deg,
scale = "row",
cluster_rows = TRUE,
cluster_cols = TRUE,
main = "RPKM Heatmap of Differentially Expressed Genes"
)
```



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```
# Quasi-likelihood method

group=c(rep("conA",2),rep("conB",2))
group
```

```
## [1] "conA" "conA" "conB" "conB"
```

```
#DGEList is an edgeR object that stores your RNA-seq count data along with information
about the experim
y=DGEList(counts=count_matrix,group=group)
keep=filterByExpr(y)
y=y[keep,,keep.lib.sizes=FALSE]
dim(y)
```

```
## [1] 5224    4
```

```
y <- calcNormFactors(y)
design=model.matrix(~0+group)
design
```

```
##   groupconA groupconB
## 1         1         0
## 2         1         0
## 3         0         1
## 4         0         1
## attr(,"assign")
## [1] 1 1
## attr(,"contrasts")
## attr(,"contrasts")$group
## [1] "contr.treatment"
```

```
# This fits a quasi-likelihood negative binomial generalized linear model to your RNA-s
eq count data.
y_glm=glmQLFit(y,design)
BvsA=makeContrasts(groupconB - groupconA, levels=design)
#This performs the actual quasi-likelihood F-test for the contrast:
res = glmQLFTest(y_glm, contrast = BvsA)
#topTags(res)

Ftest_table <- res$table
up_Ftest <- subset(Ftest_table,logFC >= 1 & PValue <= 0.01)
down_Ftest <- subset(Ftest_table,logFC <= -1 & PValue <= 0.01)
nrow(up_Ftest)
```

```
## [1] 255
```

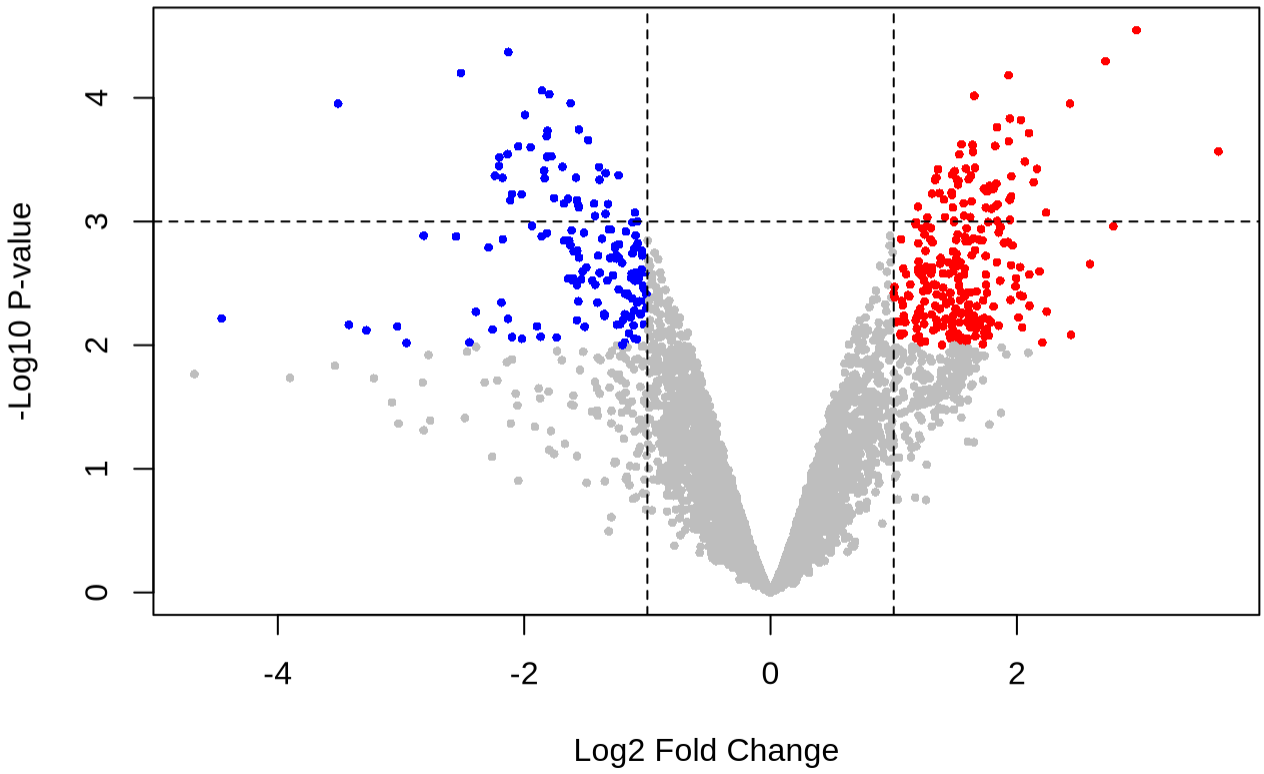
```
nrow(down_Ftest)
```

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```
## [1] 148
```

```
Ftest_table$color <- "grey"
# Upregulated genes
Ftest_table$color[Ftest_table$logFC >= 1 & Ftest_table$PValue <= 0.01] <- "red"
# Downregulated genes
Ftest_table$color[Ftest_table$logFC <= -1 & Ftest_table$PValue <= 0.01] <- "blue"
# Calculate -log10(P-value)
Ftest_table$negLogP <- -log10(Ftest_table$PValue)
# Volcano plot
plot(
Ftest_table$logFC,
Ftest_table$negLogP,
pch = 16,
cex = 0.6,
col = Ftest_table$color,
xlab = "Log2 Fold Change",
ylab = "-Log10 P-value",
main = "Volcano Plot: conB vs conA"
)
# Vertical cutoff lines
abline(
v = c(-1, 1),
lty = 2,
col = "black"
)
# Horizontal cutoff line
abline(
h = -log10(0.001),
lty = 2,
col = "black"
)
```

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```
#Load the gprofiler2 package for Gene Ontology (GO) enrichment analysis
library(gprofiler2)

#Convert the upregulated gene table into a data frame
up_classic=data.frame(up_et)

#Add the gene IDs from the row names as a separate column
up_classic$Genes=rownames(up_classic)
head(up_classic$Genes)
```

```
## [1] "YAL025C" "YBL042C" "YBL039C" "YBL024W" "YBL004W" "YBR031W"
```

```
# Remove the row names from the data frame
row.names(up_classic)=NULL

# Convert the downregulated gene table into a data frame
down_classic=data.frame(down_et)

# Add the gene IDs from the row names as a separate column
down_classic$Genes=rownames(down_classic)
head(down_classic$Genes)
```

```
## [1] "YAL008W"   "YBL064C"   "YBL015W"   "YBR053C"   "YBR072W"   "YBR089C-A"
```

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```
#Remove the row names from the data frame
row.names(down_classic)=NULL
write.table(up_classic, file="UP_regulated.csv",row.names = FALSE)
```

```
# We have to perform GO enrichment analysis for the upregulated genes organism = “scere
visiae” specifies Saccharomyces cerevisiae (yeast) Bonferroni correction is used to cor
rect for multiple testing

# Run functional profiling on upregulated genes
res_up <- gost(
  query = up_classic$Genes,
  organism = "scerevisiae",
  exclude_iea = TRUE,
  correction_method = "bonferroni"
)

# View the top enrichment results
head(res_up$result)
```

```
##     query significant       p_value term_size query_size intersection_size
## 1 query_1        TRUE 8.052105e-122       396        235               145
## 2 query_1        TRUE 2.274815e-109       488        235               147
## 3 query_1        TRUE  3.958814e-97       348        235               123
## 4 query_1        TRUE  1.459798e-95       287        235               114
## 5 query_1        TRUE  3.176909e-73       565        235               127
## 6 query_1        TRUE  1.504631e-62       138        235                69
##   precision    recall    term_id source                          term_name
## 1 0.6170213 0.3661616 GO:0042254  GO:BP                 ribosome biogenesis
## 2 0.6255319 0.3012295 GO:0022613  GO:BP ribonucleoprotein complex biogenesis
## 3 0.5234043 0.3534483 GO:0016072  GO:BP              rRNA metabolic process
## 4 0.4851064 0.3972125 GO:0006364  GO:BP                      rRNA processing
## 5 0.5404255 0.2247788 GO:0006396  GO:BP                       RNA processing
## 6 0.2936170 0.5000000 GO:0042273  GO:BP   ribosomal large subunit biogenesis
##   effective_domain_size source_order                           parents
## 1                  6547         9411                        GO:0022613
## 2                  6547         6185                        GO:0044085
## 3                  6547         4856                        GO:0016070
## 4                  6547         2032 GO:0006396, GO:0016072, GO:0042254
## 5                  6547         2049 GO:0010467, GO:0032774, GO:0044238
## 6                  6547         9421            GO:0022613, GO:0042254
```

```
# We have to extract the GO enrichment results into a data frame
df_GO=res_up$result
head(df_GO)
```

```
##     query significant       p_value term_size query_size intersection_size
## 1 query_1        TRUE 8.052105e-122       396        235               145
## 2 query_1        TRUE 2.274815e-109       488        235               147
## 3 query_1        TRUE  3.958814e-97       348        235               123
## 4 query_1        TRUE  1.459798e-95       287        235               114
## 5 query_1        TRUE  3.176909e-73       565        235               127
```

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```
## 6 query_1        TRUE  1.504631e-62       138        235                69
##   precision    recall    term_id source                          term_name
## 1 0.6170213 0.3661616 GO:0042254  GO:BP                  ribosome biogenesis
## 2 0.6255319 0.3012295 GO:0022613  GO:BP ribonucleoprotein complex biogenesis
## 3 0.5234043 0.3534483 GO:0016072  GO:BP               rRNA metabolic process
## 4 0.4851064 0.3972125 GO:0006364  GO:BP                     rRNA processing
## 5 0.5404255 0.2247788 GO:0006396  GO:BP                      RNA processing
## 6 0.2936170 0.5000000 GO:0042273  GO:BP   ribosomal large subunit biogenesis
##   effective_domain_size source_order                           parents
## 1                  6547         9411                        GO:0022613
## 2                  6547         6185                        GO:0044085
## 3                  6547         4856                        GO:0016070
## 4                  6547         2032 GO:0006396, GO:0016072, GO:0042254
## 5                  6547         2049 GO:0010467, GO:0032774, GO:0044238
## 6                  6547         9421             GO:0022613, GO:0042254
```

```
# Select the first 10 GO terms from the enrichment results
go_up_top10 <- df_GO[1:10, ]
head(go_up_top10)
```

```
##     query significant       p_value term_size query_size intersection_size
## 1 query_1        TRUE 8.052105e-122       396        235               145
## 2 query_1        TRUE 2.274815e-109       488        235               147
## 3 query_1        TRUE  3.958814e-97       348        235               123
## 4 query_1        TRUE  1.459798e-95       287        235               114
## 5 query_1        TRUE  3.176909e-73       565        235               127
## 6 query_1        TRUE  1.504631e-62       138        235                69
##   precision    recall    term_id source                          term_name
## 1 0.6170213 0.3661616 GO:0042254  GO:BP                  ribosome biogenesis
## 2 0.6255319 0.3012295 GO:0022613  GO:BP ribonucleoprotein complex biogenesis
## 3 0.5234043 0.3534483 GO:0016072  GO:BP               rRNA metabolic process
## 4 0.4851064 0.3972125 GO:0006364  GO:BP                     rRNA processing
## 5 0.5404255 0.2247788 GO:0006396  GO:BP                      RNA processing
## 6 0.2936170 0.5000000 GO:0042273  GO:BP   ribosomal large subunit biogenesis
##   effective_domain_size source_order                           parents
## 1                  6547         9411                        GO:0022613
## 2                  6547         6185                        GO:0044085
## 3                  6547         4856                        GO:0016070
## 4                  6547         2032 GO:0006396, GO:0016072, GO:0042254
## 5                  6547         2049 GO:0010467, GO:0032774, GO:0044238
## 6                  6547         9421             GO:0022613, GO:0042254
```

```
# We have to convert GO P-values to -log10(P-value) Smaller P-values will have larger -log10(P-value) values
go_up_top10$negLogP <- -log10(go_up_top10$p_value)

#Sort the GO terms according to their P-value .The most statistically significant terms come first
go_up_top10 <- go_up_top10[order(go_up_top10$p_value),]
```

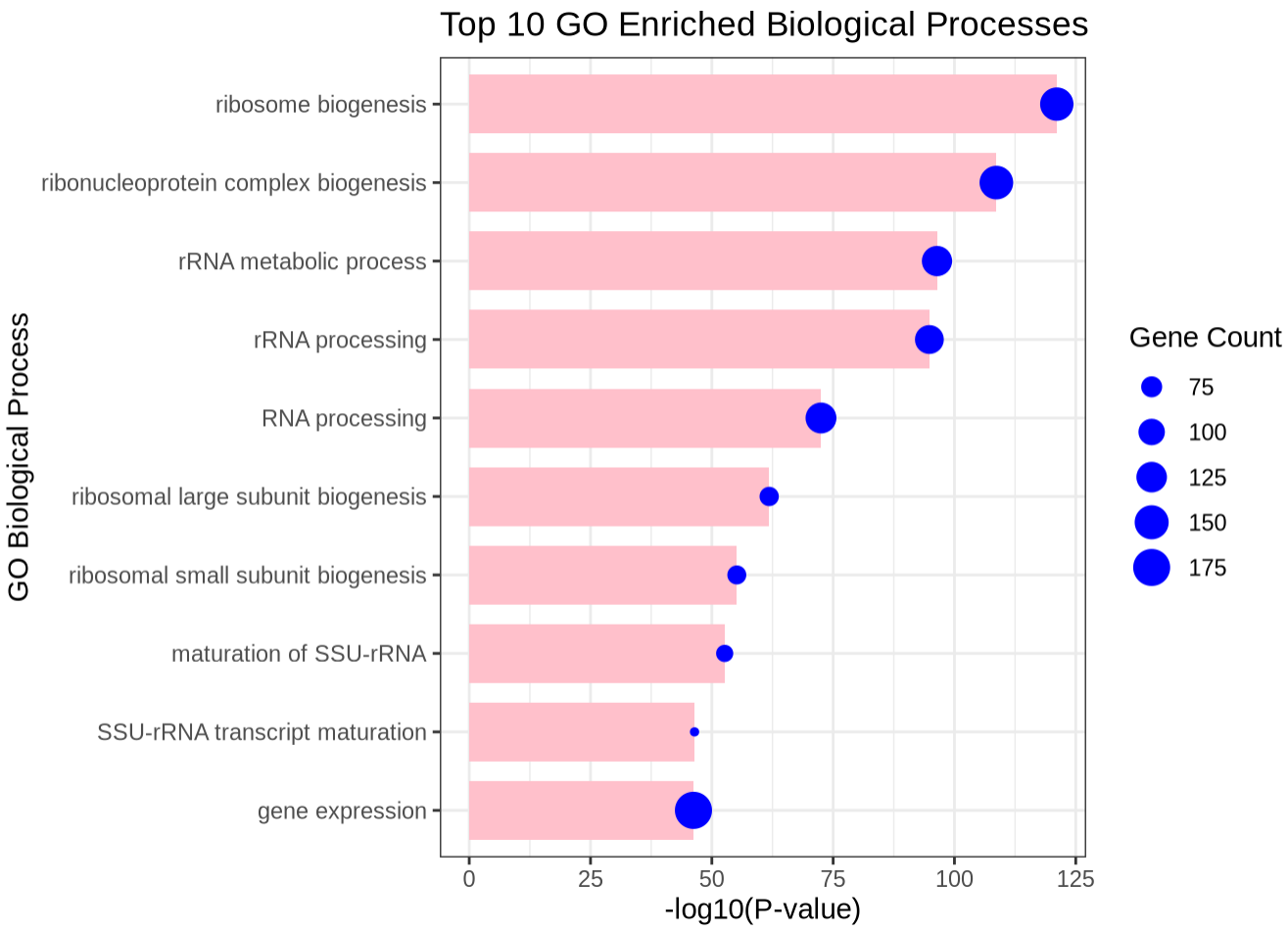
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```
#Select top 10 after sorting
top_10 <- go_up_top10[1:10, ]
```

```
# Rename the long GO term for better visualization
top_10$term_name[top_10$term_name ==
"maturation of SSU-rRNA from tricistronic rRNA transcript (SSU-rRNA, 5.8S rRNA, LSU-rRNA)"] <-
"SSU-rRNA transcript maturation"
```

```
# We have to plot showing the top 10 enriched GO terms
ggplot(top_10, aes(
x = negLogP,
y = reorder(term_name, negLogP)
)) +
geom_col(
fill = "pink",
width = 0.75
) +
geom_point(
aes(size = intersection_size),
shape = 21,
fill = "blue",
color = "blue"
) +
labs(
title = "Top 10 GO Enriched Biological Processes",
x = "-log10(P-value)",
y = "GO Biological Process",
size = "Gene Count"
) +
theme_bw()
```

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```
# Figure Description & Interpretation

# The x-axis displays statistical significance using −log10(P-value), meaning points fu
rther to the right carry higher statistical confidence. Pink bars indicate the overall
enrichment level for each GO term, while the blue circles show the number of upregulate
d genes mapped to that term, where larger circles represent a higher gene count.

# Functional enrichment analysis reveals that genes upregulated in conB relative to con
A are heavily focused on ribosome assembly and RNA processing. Ribosome biogenesis stan
ds out as the single most enriched biological process, along with ribonucleoprotein com
plex assembly, rRNA metabolism, and pre-rRNA processing. Both small and large ribosomal
subunit maturation terms are also strongly represented. Overall, these results indicate
that switching from conA to conB primarily drives cellular investment into building rib
osomes, processing rRNA, and scaling up the core translational machinery.
```

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